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Communication device

Abstract

A communication device may include a first type of interface and a second type of interface. The communication device may execute the communication of object data with a mobile device using the second type of interface after executing a specific process for causing the communication device to shift to a communication-enabled state, in a case where it is determined that the communication device is not currently in the communication-enabled state. Also, the communication device may execute the communication of the object data with the mobile device using the second type of interface without executing the specific process, in a case where it is determined that the communication device is currently in the communication-enabled state.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of prior U.S. application Ser. No. 18/164,784, filed on Feb. 6, 2023, which is a continuation of prior U.S. application Ser. No. 17/318,630, filed on May 12, 2021, now U.S. Pat. No. 11,582,592 B2, issued May 12, 2021, which is a continuation of prior U.S. application Ser. No. 16/884,953, filed on May 27, 2020, now U.S. Pat. No. 11,012,843 B2, issued May 18, 2021, which is a continuation of prior U.S. application Ser. No. 16/560,686, filed Sep. 4, 2019, now U.S. Pat. No. 10,674,341 B2, issued Jun. 2, 2020, which is a continuation of prior U.S. application Ser. No. 16/165,731, filed Oct. 19, 2018, now U.S. Pat. No. 10,492,051 B2, issued Nov. 26, 2019, which is a continuation of prior U.S. application Ser. No. 15/473,130, filed Mar. 29, 2017, now U.S. Pat. No. 10,123,193 B2, issued Nov. 6, 2018, which is a continuation of prior U.S. application Ser. No. 14/789,644, filed Jul. 1, 2015, now U.S. Pat. No. 9,973,914 B2, issued May 15, 2018, which is a continuation of prior U.S. application Ser. No. 13/832,035, filed Mar. 15, 2013, now U.S. Pat. No. 9,100,774 B2, issued Aug. 4, 2015, which claims priority to Japanese Patent Application No. 2012-082817, filed on Mar. 30, 2012, the contents of which are hereby incorporated by reference into the present application.

TECHNICAL FIELD

(1) A technique disclosed in the present specification relates to a communication device for executing communication of object data with a mobile device.

DESCRIPTION OF RELATED ART

(2) A technique for two communication devices to execute wireless communication is known. The two communication devices execute communication of a wireless setting according to a short-range wireless communication system (i.e., a wireless communication according to NFC (abbreviation of: Near Field Communication)). The wireless setting is a setting for executing wireless communication according to a communication system different from the NFC system (e.g., IEEE 802.11a, 802.11b). Thereby, the two communication devices become capable of executing wireless communication according to the wireless setting.

SUMMARY

(3) The present specification discloses a technique for a communication device to appropriately execute communication with a mobile device.

(4) A technique disclosed herein is a communication device. The communication device may comprise a first type of interface for executing a communication with a portable terminal and a second type of interface for executing a communication with the mobile device. The communication device may comprise one or more processors and a memory that stores computer-readable instructions therein. The computer-readable instructions, when executed by the one or more processors, may cause the communication device to execute (A) receiving specific information from the mobile device using the first type of interface, (B) determining, using the specific information, whether or not the communication device is currently in a communication-enabled state in which the communication device is currently capable of executing a communication of object data with the mobile device using the second type of interface, (C) executing the communication of the object data with the mobile device using the second type of interface after executing a specific process for causing the communication device to shift to the communication-enabled state, in a case where it is determined that the communication device is not currently in the communication-enabled state and (D) executing the communication of the object data with the mobile device using the second type of interface without executing the specific process, in a case where it is determined that the communication device is currently in the communication-enabled state.

(5) Moreover, a control method, a computer program, and a non-transitory computer-readable storage medium computer-readable instructions for the communication device, are also novel and

useful. Further, a communication system including the communication device and the mobile device are also novel and useful.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** shows the configuration of a communication system.
(2) FIG. **2** shows a flowchart of a communication process executed by a multi-function peripheral of a first embodiment.
(3) FIG. **3** shows a sequence view for explaining processes executed by devices in a first situation.
(4) FIG. **4** shows a sequence view for explaining processes executed by devices in a second situation.
(5) FIG. **5** shows a sequence view for explaining processes executed by devices in a third situation.
(6) FIG. **6** shows a sequence view for explaining processes executed by devices in a fourth situation.
(7) FIG. **7** shows a sequence view for explaining processes executed by devices in a fifth situation.
(8) FIG. **8** shows a flowchart of a communication process executed by a multi-function peripheral of a second embodiment.
(9) FIG. **9** shows a sequence view for explaining processes executed by devices in a sixth situation.
(10) FIG. **10** shows a sequence view for explaining processes executed by devices in a seventh situation.
(11) FIG. **11** shows a flowchart of a communication process executed by a multi-function peripheral of a third embodiment.
(12) FIG. **12** shows a sequence view for explaining processes executed by devices in an eighth situation.
(13) FIG. **13** shows a sequence view for explaining processes executed by devices in a ninth situation.
(14) FIG. **14** shows a flowchart of a communication process executed by a multi-function peripheral of a fourth embodiment.
(15) FIG. **15** shows a sequence view for explaining processes executed by devices in a tenth situation.
(16) FIG. **16** shows a sequence view for explaining processes executed by devices in an eleventh situation.
(17) FIG. **17** shows a flowchart of a communication process executed by a multi-function peripheral of a fifth embodiment.
(18) FIG. **18** shows a sequence view for explaining processes executed by devices in a twelfth situation.
(19) FIG. **19** shows a flowchart of a communication process executed by a multi-function peripheral of a sixth embodiment.
(20) FIG. **20** shows a flowchart of a communication process executed by a multi-function peripheral of a seventh embodiment.
(21) FIG. **21** shows a sequence view for explaining processes executed by devices in the seventh embodiment.
DETAILED DESCRIPTION (EMBODIMENT)
First Embodiment
(22) (Configuration of Communication System)
(23) As shown in FIG. **1**, a communication system **2** comprises a multi-function peripheral (called “MFP” (abbreviation of. Multi-Function Peripheral) below) **10**, a mobile device **50**, an access point (called “AP” below) **6**, and a PC **8**. The MFP **10** and the mobile device **50** are capable of executing

short-range wireless communication. The short-range wireless communication is according to the wireless communication NFC system. In the present embodiment, the wireless communication is executed according to the NFC system based on international standards ISO/IEC 21481 or 18092.
(24) Further, the MFP **10** is capable of executing wireless communication according to the Wi-Fi Direct system (to be described). Below, Wi-Fi Direct is called “WFD”. In WFD, wireless communication is executed based on IEEE (abbreviation of. The Institute of Electrical and Electronics Engineers, Inc.) 802.11 standard and standards based on thereon (e.g., 802.11a, 11b, 11g, 11n, etc.). The NFC system and the system of WFD (called “WFD system” below) have different wireless communication systems (i.e., wireless communication standards). Further, the communication speed of wireless communication according to the WFD system is faster than the communication speed of wireless communication according to the NFC system.
(25) For example, the MFP **10** can construct a WFD network by establishing a connection with the mobile device **50** according to the WFD system (called “WFD connection” below). Similarly, the MFP **10** can construct a WFD network by establishing a WFD connection with the PC **8**.
(26) The PC **8**, the MFP **10** and the mobile device **50** are further capable of executing wireless communication according to a normal Wi-Fi system (e.g., IEEE 802.11) different from the WFD system. In general terms, wireless communication according to normal Wi-Fi is wireless communication using the AP **6**, and wireless communication according to the WFD system is wireless communication not using the AP **6**. For example, the MFP **10** can belong to a normal Wi-Fi network by establishing a connection with the AP **6** (called “normal Wi-Fi connection” below) according to normal Wi-Fi. Via the AP **6**, the MFP **10** can execute wireless communication with another device belonging to the normal Wi-Fi network (e.g., the PC **8**, the mobile device **50**). Moreover, the NFC system and the system of normal Wi-Fi (called “the normal Wi-Fi system” below) have different wireless communication systems (i.e., wireless communication standards). Further, the communication speed of normal Wi-Fi is faster than the communication speed of NFC.
(27) (WFD)
(28) WFD is a standard formulated by Wi-Fi Alliance. WFD is described in “Wi-Fi Peer-to-Peer (P2P) Technical Specification Version 1.1”, created by Wi-Fi Alliance.
(29) As described above, the PC **8**, the MFP **10**, and the mobile device **50** are each capable of executing wireless communication according to the WFD system. Below, an apparatus capable of executing wireless communication according to the WFD system is called a “WFD-compatible apparatus”. According to the WFD standard, three states are defined as the states of the WFD-compatible apparatus: Group Owner state (called “G/O state” below), client state, and device state. The WFD-compatible apparatus is capable of selectively operating in one state among the three states.
(30) One WFD network includes an apparatus in the G/O state and an apparatus in the client state. Only one G/O state apparatus can be present in the WFD network, but one or more client state apparatuses can be present. The G/O state apparatus manages the one or more client state apparatuses. Specifically, the G/O state apparatus creates an administration list in which identification information (i.e., MAC address) of each of the one or more client state apparatuses is written. When a client state apparatus newly belongs to the WFD network, the G/O state apparatus adds the identification information of that apparatus to the administration list, and when the client state apparatus leaves the WFD network, the G/O state apparatus deletes the identification information of that apparatus from the administration list.
(31) The G/O state apparatus is capable of wirelessly communicating object data (e.g., data that includes network layer information of the OSI reference model (print data, scan data, etc.)) with an apparatus registered in the administration list, i.e., with a client state apparatus (i.e., an apparatus belonging to the WFD network). However, with an unregistered apparatus which is not registered in the administration list, the G/O state apparatus is capable of wirelessly communicating data for the unregistered apparatus to belong to the WFD network (e.g., data that does not include network

layer information (physical layer data such as a Probe Request signal, Probe Response signal, etc.)), but is not capable of wirelessly communicating the object data. For example, the MFP **10** that is in the G/O state is capable of wirelessly receiving print data from the mobile device **50** that is registered in the administration list (i.e., the mobile device **50** that is in the client state), but is not capable of wirelessly receiving print data from an apparatus that is not registered in the administration list.

(32) Further, the G/O state apparatus is capable of relaying the wireless communication of object data (print data, scan data, etc.) between a plurality of client state apparatuses. For example, in a case where the mobile device **50** that is in the client state is to wirelessly send print data to another printer that is in the client state, the mobile device **50** first wirelessly sends the print data to the MFP **10** that is in the G/O state. In this case, the MFP **10** wirelessly receives the print data from the mobile device **50**, and wirelessly sends the print data to the other printer. That is, the G/O state apparatus is capable of executing the function of an AP of the normal wireless network.

(33) Moreover, a WFD-compatible apparatus that does not belong to the WFD network (i.e., an apparatus not registered in the administration list) is a device state apparatus. The device state apparatus is capable of wirelessly communicating data for belonging to the WFD network (physical layer data such as a Probe Request signal, Probe Response signal, etc.), but is not capable of wirelessly communicating object data (print data, scan data, etc.) via the WFD network.

(34) Moreover, below, an apparatus that is not capable of executing wireless communication according to the WFD system, but is capable of executing wireless communication according to normal Wi-Fi is called a "WFD-incompatible apparatus". The "WFD-incompatible apparatus" may also be called a "legacy apparatus". A WFD-incompatible apparatus cannot operate in the G/O state. A G/O state apparatus can register identification information of the WFD-incompatible apparatus in the administration list.

(35) (Configuration of MFP **10**)

(36) The MFP **10** comprises an operating unit **12**, a displaying unit **14**, a print executing unit **16**, a scan executing unit **18**, a wireless LAN interface (an "interface" is called "I/F" below) **20**, an NFC I/F **22**, and a control unit **30**. The operating unit **12** consists of a plurality of keys. A user can input various instructions to the MFP **10** by operating the operating unit **12**. The displaying unit **14** is a display for displaying various types of information. The print executing unit **16** is an ink jet system, laser system, etc. printing mechanism. The scan executing unit **18** is a CCD, CIS, etc. scanning mechanism.

(37) The wireless LAN I/F **20** is an interface for the control unit **30** to execute wireless communication according to the WFD system and wireless communication according to normal Wi-Fi. The wireless LAN I/F **20** is physically one interface. However, a MAC address used in wireless communication according to the WFD system (called "MAC address for WFD" below) and a MAC address used in wireless communication according to normal Wi-Fi (called "MAC address for normal Wi-Fi" below) are both assigned to the wireless LAN I/F **20**. More specifically, the MAC address for the normal Wi-Fi is pre-assigned to the wireless LAN I/F **20**. Using the MAC address for the normal Wi-Fi, the control unit **30** creates the MAC address for WFD, and assigns the MAC address for WFD to the wireless LAN I/F **20**. The MAC address for WFD differs from the MAC address for the normal Wi-Fi. Consequently, via the wireless LAN I/F **20**, the control unit **30** can simultaneously execute both wireless communication according to the WFD system and wireless communication according to the normal Wi-Fi. Consequently, a situation can be established in which the MFP **10** belongs to the WFD network and belongs to the normal Wi-Fi network.

(38) Moreover, the G/O state apparatus can write, in the administration list, not only the identification information of the WFD-compatible apparatus that is in the client state, but also the identification information of a WFD-incompatible apparatus. That is, the G/O state apparatus can also establish the WFD connection with the WFD-incompatible apparatus. In general terms, the

WFD connection is a wireless connection in which the MAC address for the WFD of the MFP **10** is used. Further, the WFD network is a wireless network in which the MAC address for the WFD of the MFP **10** is used. Similarly, the normal Wi-Fi connection is a wireless connection in which the MAC address for the normal Wi-Fi of the MFP **10** is used. Further, the normal Wi-Fi network is a wireless network in which the MAC address for the normal Wi-Fi of the MFP **10** is used.

(39) By operating the operating unit **12**, the user can change a setting of the wireless LAN I/F **20**, thereby being able to change to either mode of a mode in which wireless communication according to the WFD system using the wireless LAN I/F **20** can be executed (called "WFD=ON mode" below), and a mode in which wireless communication according to the WFD system using the wireless LAN I/F **20** cannot be executed (called "WFD=OFF mode" below). A mode setting unit **46** sets the mode to either WFD=ON mode or WFD=OFF mode in accordance with the operation of the user. Specifically, the mode setting unit **46** stores, in the memory **34**, a mode value representing the mode set by the user.

(40) Moreover, in the WFD I/F=OFF mode state, the control unit **30** cannot execute processes according to the WFD system (e.g., a process of setting the MFP **10** to spontaneous G/O mode (to be described), G/O negotiation, etc.). In the WFD I/F=ON state, the memory **34** stores values representing the current state of the MFP **10** relating to WFD (the state from among G/O state, client state, and device state).

(41) The NFC I/F **22** is an interface for the control unit **30** to execute wireless communication according to the NFC system. The NFC I/F **22** is formed of a chip differing physically from the wireless LAN I/F **20**.

(42) Moreover, the communication speed of wireless communication via the wireless LAN I/F **20** (e.g., maximum communication speed is 11 to 454 Mbps) is faster than the communication speed of wireless communication via the NFC I/F **22** (e.g., maximum communication speed is 100 to 424 Kbps). Further, the frequency of the carrier wave in wireless communication via the wireless LAN I/F **20** (e.g., 2.4 GHz band, 5.0 GHz band) differs from the frequency of the carrier wave in the wireless communication via the NFC I/F **22** (e.g., 13.56 MHz band). Further, in a case where the distance between the MFP **10** and the mobile device **50** is less than or equal to approximately 10 cm, the control unit **30** can wirelessly communicate with the mobile device **50** according to the NFC system via the NFC I/F **22**. In a case where the distance between the MFP **10** and the mobile device **50** is either less than or equal to 10 cm, or is greater than or equal to 10 cm (e.g., a maximum is approximately 100 m), the control unit **30** can wirelessly communicate, via the wireless LAN I/F **20**, with the mobile device **50** according to the WFD system and according to the normal Wi-Fi. That is, the maximum distance across which the MFP **10** can execute wireless communication with a communication destination apparatus (e.g., the mobile device **50**) via the wireless LAN I/F **20** is greater than the maximum distance across which the MFP **10** can execute the wireless communication with the communication destination apparatus via the NFC I/F **22**.

(43) The control unit **30** comprises a CPU **32** and the memory **34**. The CPU **32** executes various processes according to programs stored in the memory **34**. The CPU **32** realizes the functions of the units **40** to **46** by executing processes according to the programs.

(44) The memory **34** is formed of a ROM, RAM, hard disk, etc. The memory **34** stores the programs executed by the CPU **32**. The memory **34** comprises a work area **38**. In the case where the MFP **10** currently belongs to a WFD network, the work area **38** stores information indicating that the MFP **10** currently belongs to the WFD network, and a wireless setting (including authentication method, encryption method, password, SSID (Service Set Identifier) and BSSID (Basic Service Set Identifier) of the wireless network) for communicating object data (e.g., print data) via the WFD network. Further, in the case where the MFP **10** currently belongs to a normal Wi-Fi network, the work area **38** stores information indicating that the MFP **10** currently belongs to the normal Wi-Fi network, and a wireless setting for communicating object data via the normal Wi-Fi network. The SSID of the WFD network is a network identifier for identifying the WFD

network, and the SSID of the normal Wi-Fi network is a network identifier for identifying the normal Wi-Fi network. The BSSID of the WFD network is an identifier unique to the G/O state apparatus (e.g., the MAC address of the G/O state apparatus), and the BSSID of the normal Wi-Fi network is an identifier unique to the AP (e.g., a unique identifier of the AP).

(45) In the case where the MFP **10** is operating according to the WFD system, the work area **38** further stores a value indicating the current state of WFD (one state from among G/O state, client state, or device state). The work area **38** further stores a mode value representing the WFD=ON mode, or a mode value representing the WFD=OFF mode.

(46) Moreover, by operating the operating unit **12**, the user can set the MFP **10** to spontaneous G/O mode. Spontaneous G/O mode is a mode for maintaining the operation of the MFP **10** in the G/O state. The work area **38** within the memory **34** further stores a value indicating whether the MFP **10** has been set to spontaneous G/O mode. When the WFD-compatible apparatus that is in the device state is to establish a WFD connection with another WFD-compatible apparatus that is in the device state, the WFD-compatible apparatus usually executes G/O negotiation to selectively determine which state, of G/O state and client state, it is to operate in. In the case where the MFP **10** has been set to the spontaneous G/O mode, the MFP **10** maintains operation in the G/O state without executing G/O negotiation.

(47) (Configuration of Mobile Device **50**)

(48) The mobile device **50** is, for example, a mobile phone (e.g., a Smart Phone), PDA, notebook PC, tablet PC, portable music player, portable video player, etc. The mobile device **50** comprises two wireless interfaces, a wireless LAN I/F (i.e., an interface for WFD and normal Wi-Fi) and an NFC I/F. Consequently, the mobile device **50** is capable of executing wireless communication with the MFP **10** using the wireless LAN I/F, and is capable of executing wireless communication with the MFP **10** using the NFC I/F. The mobile device **50** comprises an application program for causing the MFP **10** to execute functions (e.g., print function, scan function, etc.). Moreover, the application program may, for example, be installed on the mobile device **50** from a server provided by a vendor of the MFP **10**, or may be installed on the mobile device **50** from a media shipped together with the MFP **10**.

(49) Like the MFP **10**, the mobile device **50** comprises a work area **58** within a memory **54**. In the case where the mobile device **50** currently belongs to the WFD network or the normal Wi-Fi network, the work area **58** stores a wireless setting (including authentication method, encryption method, password, SSID and BSSID of the wireless network) for executing communication via the relevant network. Further, in the case where the mobile device **50** is operating according to the WFD system, the work area **58** stores a state value representing the state of the mobile device **50** (i.e., one state from among G/O state, client state and device state).

(50) (Configuration of PC **8**)

(51) The PC **8** comprises a wireless LAN I/F (i.e., an interface for WFD and normal Wi-Fi), but does not comprise an NFC I/F. Consequently, the PC **8** is capable of executing communication with the MFP **10** by using the wireless LAN I/F, but is not capable of executing wireless communication according to the NFC system. The PC **8** comprises a driver program for causing the MFP **10** to execute a process (e.g., print process, scan process, etc.). Moreover, the driver program is usually installed on the PC **8** from a media shipped together with the MFP **10**. However, in a modification, the driver program may be installed on the PC **8** from a server provided by the vendor of the MFP **10**.

(52) (Configuration of AP **6**)

(53) The AP **6** is not a WFD G/O state apparatus, but is a standard access point called a wireless access point or wireless LAN router. The AP **6** can establish a normal Wi-Fi connection with a plurality of apparatuses. Thereby, a normal Wi-Fi network including the AP **6** and the plurality of apparatuses is constructed. The AP **6** receives data from one apparatus from among the plurality of apparatuses belonging to the normal Wi-Fi network, and sends the data to another one apparatus

from among the plurality of apparatuses. That is, the AP **6** relays communication between a pair of apparatuses belonging to the normal Wi-Fi network.
(54) Moreover, differences between the WFD G/O state apparatus and the normal AP are as follows. In the case where the WFD G/O state apparatus disconnects from the WFD network to which it currently belongs, and newly belongs to another WFD network, the WFD G/O state apparatus can operate in a state other than the G/O state (i.e., the client state). By contrast, a normal AP (i.e., the AP **6**) executes the function of relaying communication between the pair of apparatuses regardless of which normal Wi-Fi network the normal AP belongs to, and the normal AP cannot operate in the client state.
(55) (Communication Process Executed by MFP **10**)
(56) A communication process executed by the MFP **10** will be described with reference to FIG. **2**. When a power source of the MFP **10** is turned ON, the control unit **30** executes a communication process. In S**2**, a receiving unit **40** monitors whether NFC information has been received by executing wireless communication according to the NFC system. Moreover, the receiving unit **40** receives the NFC information via the NFC I/F **22**. Specifically, the receiving unit **40** monitors whether an NFC communication session has been established between the MFP **10** and the mobile device **50**. While the power source of the MFP **10** is ON, the receiving unit **40** causes the NFC I/F **22** to transmit radio waves for detecting a device capable of executing wireless communication according to the NFC system.
(57) The user of the mobile device **50** activates the application program. By operating the mobile device **50**, the user causes the mobile device **50** to create NFC information that includes a process execution instruction (e.g., print instruction, scan instruction) indicating a process that the MFP **10** is to execute. In the case where the mobile device **50** currently belongs to a wireless network, the NFC information further includes the SSID and BSSID of the wireless network to which the mobile device **50** currently belongs. Moreover, the case where the mobile device **50** currently belongs to the wireless network is a case in which a wireless connection, this being the WFD connection or the normal Wi-Fi connection, or both, has been established between the mobile device **50** and another device (e.g., the AP **6**, the MFP **10**).
(58) The user can bring the mobile device **50** closer to the MFP **10**. Thereby, when the distance between the mobile device **50** and the MFP **10** becomes less than the distance (e.g., 10 cm) where the radio waves reach each other, the mobile device **50** receives a radio wave from the MFP **10**, and sends a response wave to the MFP **10**. Consequently, the control unit **30** receives the response wave from the mobile device **50**, and an NFC communication session is established. When the NFC communication session has been established, the mobile device **50** sends the created NFC information to the MFP **10**.
(59) Upon receiving the NFC information (YES in S**2**), in S**4** a determining unit **42** determines whether the MFP **10** currently belongs to a network. Specifically, in the case where the work area **38** stores information indicating that the MFP **10** currently belongs to the WFD network or information indicating that the MFP **10** currently belongs to the normal Wi-Fi network, or both, the determining unit **42** determines that the MFP **10** currently belongs to a wireless network (YES in S**4**), and the process proceeds to S**6**. On the other hand, in the case where neither the information indicating that the MFP **10** currently belongs to the WFD network or the information indicating that the MFP **10** currently belongs to the normal Wi-Fi network is stored in the work area **38**, the determining unit **42** determines that the MFP **10** does not currently belong to a wireless network (NO in S**4**), and the process proceeds to S**8**.
(60) In S**6**, the determining unit **42** confirms whether the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs. Specifically, the determining unit **42** first determines whether the SSID and BSSID of the network to which the mobile device **50** currently belongs are included in the NFC information. In the case where the SSID and BSSID are not included in the NFC information, the determining unit **42** determines that the mobile device **50**

does not currently belong to the network to which the MFP **10** currently belongs (NO in S**6**). According to this configuration, the MFP **10** can appropriately determine that the mobile device **50** does not currently belong to the network to which the MFP **10** currently belongs. In the case where the SSID and BSSID of the network to which the mobile device **50** currently belongs are included in the NFC information, the determining unit **42** determines whether the SSID and BSSID included in the wireless setting stored in the work area **38** are identical to the SSID and BSSID included in the NFC information.
(61) In the case where the SSIDs and BSSIDs are both identical, it is determined that the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs (YES in S**6**), and the process proceeds to S**7**. On the other hand, in the case where the SSIDs or the BSSIDs, or both are not identical, it is determined that the mobile device **50** does not currently belong to the network to which the MFP **10** currently belongs (NO in S**6**), and the process proceeds to S**8**. According to this configuration, the MFP **10** can appropriately determine whether the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs. Moreover, in S**6** the determining unit **42** determines whether the SSIDs are identical, and whether the BSSIDs are identical. Thereby, the determining unit **42** can determine whether the MFP **10** and the mobile device **50** belong to the same wireless network constructed by the same AP. More specifically, one AP may construct a plurality of wireless networks by using a plurality of SSID s. Consequently, in the case where the BSSIDs are identical and the SSIDs are not identical, the MFP **10** and the mobile device **50** could belong to different wireless networks constructed by the same AP. In the present embodiment, it is possible to determine more reliably whether the MFP **10** and the mobile device **50** belong to the same wireless network by determining whether both the SSIDs and BSSIDs are identical. Moreover, in a modification, it is determined in S**6** whether the SSIDs are identical, but it need not be determined whether the BSSIDs are identical. Thereby, if the SSIDs are identical, it can be determined that the MFP **10** and the mobile device **50** belong to the same wireless network even in the case where the MFP **10** and the mobile device **50** each belong to a wireless network constructed by a different access point.
(62) In the case where the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs, the MFP **10** and the mobile device **50** can execute communication via the network to which they currently belong. That is, the mobile device **50** can execute wireless communication with the MFP **10** by using the wireless setting currently stored in the work area **58**. In S**7** the control unit **30** sends, without changing the wireless setting of the mobile device **50**, information indicating setting change is unnecessary via the NFC I/F **22**, this information indicating setting change is unnecessary indicating that the communication of data can be executed, and the process proceeds to S**20**. Moreover, the information indicating setting change is unnecessary includes the IP address of the MFP **10**.
(63) In S**8** the determining unit **42** determines whether WFD=ON mode has been set. In the case where the mode value stored in the memory **34** is a value representing WFD=ON mode, the determining unit **42** determines YES in S**8**, and proceeds to S**10**. On the other hand, in the case where the mode value stored in the memory **34** is a value representing WFD=OFF mode, the determining unit **42** determines NO in S**8**, and the process proceeds to S**9**.
(64) In S**9** a communication executing unit **44** changes the mode from WFD=OFF mode to WFD=ON mode by changing the mode value stored in the memory **34**, and the process proceeds to S**15**. The communication executing unit **44** further stores, in the memory **34**, setting change information indicating that the mode value has been changed.
(65) In S**10** the determining unit **42** determines whether the MFP **10** is operating in the client state in the wireless network to which it currently belongs. Specifically, in the case where the state value stored in the work area **38** is a value representing the client state, the determining unit **42** determines that the MFP **10** is operating in the client state (YES in S**10**). On the other hand, in the case where the state value stored in the work area **38** is not a value representing the client state, the

determining unit **42** determines that the MFP **10** is not operating in the client state (NO in S**10**). In the case of YES in S**10**, the process proceeds to S**14**.
(66) On the other hand, in the case of NO in S**10**, in S**12** the determining unit **42** determines whether the MFP **10** is operating in the G/O state in the wireless network to which it currently belongs. Specifically, in the case where the state value stored in the work area **38** is a value representing the G/O state, the determining unit **42** determines that the MFP **10** is operating in the G/O state (YES in S**12**). On the other hand, in the case where the state value stored in the work area **38** is not a value representing the G/O state, the determining unit **42** determines that the MFP **10** is not operating in the G/O state (i.e., the MFP **10** is in the device state) (NO in S**12**). In the case of YES in S**12**, the process proceeds to S**13**, and in the case of NO in S**12**, the process proceeds to S**15**.
(67) In S**13** the determining unit **42** determines whether or not the number of apparatuses other than the MFP **10** included in the WFD network in which the MFP **10** is operating in the G/O state (i.e., apparatuses which have established a connection with the MFP **10**) is less than a predetermined maximum client number. The determining unit **42** determines YES in S**13** in the case where the number of identification information of apparatuses stored in the administration list is less than the maximum client number, and determines NO in S**13** in the case where the number is the same. In the case of YES in S**13**, the process proceeds to S**16**, and in the case of NO in S**13**, the process proceeds to S**14**.
(68) In S**14** the communication executing unit **44** sends communication NG information to the mobile device **50** by using the NFC I/F **22** the process returns to S**2**. This communication NG information may indicate that the MFP **10** and the mobile device **50** currently cannot execute communication.
(69) In S**15** the communication executing unit **44** sets the MFP **10** to spontaneous G/O mode. Spontaneous G/O mode is a mode which keeps the MFP **10** operating in the G/O state. Consequently, the MFP **10** is set to the G/O state although a WFD network has not been constructed at the stage of S**15**. In the case where the MFP **10** is set to the G/O state, the communication executing unit **44** prepares a wireless setting (SSID, BSSID, authentication method, encryption method, password, etc.) for the WFD-compatible apparatus and/or the WFD-incompatible apparatus to execute wireless communication, via the WFD network, with the MFP **10** that is operating in the G/O state. According to this configuration, the MFP **10** can execute wireless communication with the apparatus that receives the wireless setting from the MFP **10** regardless of whether the apparatus that receives the wireless setting (the mobile device **50** in the present embodiment) is a WFD-compatible apparatus or a WFD-incompatible apparatus.
(70) Moreover, the authentication method and encryption method are predetermined. Further, the communication executing unit **44** creates a password. Moreover, the SSID may be created by the communication executing unit **44** at the time the password is created, or may be predetermined. The BSSID is the MAC address of the MFP **10**. Moreover, at this stage, identification information of the apparatus connected with the G/O state apparatus is not described in the administration list managed by the MFP **10**.
(71) In S**16**, the communication executing unit **44** sends the prepared wireless setting to the mobile device **50** using the NFC I/F **22**. In the case where process S**16** is executed after process S**15**, the communication executing unit **44** sends, to the mobile device **50**, the wireless setting which was prepared at the stage of setting the spontaneous G/O mode (S**15**). In the case where process S**16** is executed after process S**13**, the communication executing unit **44** uses the NFC I/F **22** to send, to the mobile device **50**, the wireless setting which was prepared at the stage of constructing the WFD network in which the MFP **10** is operating in the G/O state.
(72) Next, in S**18** the communication executing unit **44** establishes a WFD connection between the MFP **10** and the mobile device **50** by using the wireless LAN I/F **20**. Upon receiving, from the MFP **10**, the wireless setting of the MFP **10** that is operating in the G/O state, the mobile device **50**

stores the received wireless setting in the work area **58**. Consequently, the mobile device **50** executes wireless communication according to normal Wi-Fi. Next, the communication executing unit **44** executes the wireless communication of an Authentication Request, Authentication Response, Association Request, Association Response, and 4-way handshake with the mobile device **50**. Various authentication processes such as authentication of SSID, authentication of authentication method and encryption method, authentication of password, etc. are executed during the course of the wireless communication. In a case where all the authentications succeed, a wireless connection is established between the MFP **10** and the mobile device **50**.

(73) Moreover, in the process S**18**, the communication executing unit **44** acquires the MAC address of the mobile device **50** by using the wireless LAN I/F **20**. When the wireless connection has been established, the control unit **30** further adds the MAC address of the mobile device **50** to the administration list. Moreover, the MAC address of the mobile device **50** is included in the NFC information. Thereby, the MFP **10** that is in the G/O state becomes able to communicate object data (print data, scan data, etc.) with the mobile device **50** according to the normal Wi-Fi. Moreover, the object data includes network layer data, which is a layer higher than the physical layer of the OSI reference model. Consequently, the MFP **10** that is in the G/O state can execute wireless communication of the network layer with the mobile device **50** that is in the client state.

(74) Next, in S**20** the communication executing unit **44** executes a data communication process with the mobile device **50** via the wireless LAN I/F **20**. The contents of the data communication process vary depending on the contents of the process execution instruction included in the NFC information. In the case where the process execution instruction is a print instruction, the communication executing unit **44** receives print data from the mobile device **50** in the data communication process. In this case, the control unit **30** causes the print executing unit **16** to execute a print process using the received print data.

(75) On the other hand, in the case where the process execution instruction is a scan instruction, the control unit **30** causes the scan executing unit **18** to scan a document that has been set on the scan executing unit **18**, creating scan data. Next, the communication executing unit **44** sends the created scan data to the mobile device **50**.

(76) Next, in S**21** the communication executing unit **44** monitors, by using the wireless LAN I/F **20**, whether a disconnection request for disconnecting the connection with the mobile device **50** has been received from the mobile device **50**. In the case where a disconnection request has not been received even though a predetermined time has elapsed (NO in S**21**), the process returns to S**2**. On the other hand, in the case where a disconnection request has been received from the mobile device **50** within a predetermined time since the ending of the data communication process of S**20** (YES in S**21**), the communication executing unit **44** disconnects the wireless connection with the mobile device **50**. Specifically, the communication executing unit **44** deletes the MAC address of the mobile device **50** within the administration list. Next, in S**22** the communication executing unit **44** determines whether the setting of the wireless LAN I/F **20** was changed by the process S**9**. Specifically, in the case where setting change information is being stored in the memory **34**, the communication executing unit **44** determines that the mode value was changed in S**9** from the mode value indicating WFD=OFF mode to the mode value indicating WFD=ON mode (YES in S**22**), and proceeds to S**23**. On the other hand, in the case where setting change information is not being stored in the memory **34**, the communication executing unit **44** determines that the mode value was not changed in S**9** from the mode value indicating WFD=OFF mode to the mode value indicating WFD=ON mode (NO in S**22**), and the process returns to S**2**.

(77) In S**23**, the communication executing unit **44** determines whether an external device (e.g., the PC **8**) other than the mobile device **50** currently belongs to the WFD network newly constructed in S**18**. Specifically, in the case where identification information other than the identification information of the mobile device **50** is included in the administration list, the communication executing unit **44** determines that the external device currently belongs to the WFD network (YES

in S**23**). In this case, without changing the mode value, the process returns to S**2**. According to this configuration, it is possible to prevent the MFP **10** from being disconnected from the WFD network in the case where the external device currently belongs to the WFD network.

(78) On the other hand, in the case where identification information other than the mobile device **50** is not included in the administration list, the communication executing unit **44** determines that the external device does not currently belong to the WFD network (NO in S**23**), and proceeds to S**24**. In S**24** the communication executing unit **44** changes the mode value from the mode value indicating WFD=ON mode to the mode value indicating the WFD=OFF mode, and the process returns to S**2**. That is, in the communication process, in the case where it is determined in S**8** that the mode value is the WFD=OFF mode, the mode value is changed from the WFD=OFF mode to the WFD=ON mode so that wireless communication with the mobile device **50** is executed temporarily via the WFD network by using the wireless LAN I/F **20**. When the mode value is changed from the WFD=ON mode to the WFD=OFF mode in S**25**, the WFD network constructed in S**18** ceases to exist. According to this configuration, in the case where the mode value was changed from the mode value indicating WFD=OFF mode to the mode value indicating the WFD=ON mode during the communication process, it is possible to return to the setting from before the mode value was changed.

Advantages of Present Embodiment

(79) Advantages of the present embodiment in first to fifth situations will be described with reference to FIGS. **3** to **7**. Moreover, processes corresponding to the communication process of FIG. **2** are shown in each of FIG. **3** to **7**.

(80) (First Situation)

(81) The first situation shown in FIG. **3** is a situation where the MFP **10** and the mobile device **50** currently belong to the same WFD network or the same normal Wi-Fi network. In this situation, when NFC information is received from the mobile device **50** by using the NFC I/F **22**, in S**6** the MFP **10** determines that the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs (YES in S**6**). In S**7** the MFP **10** sends the information indicating setting change is unnecessary to the mobile device **50** by using the NFC I/F **22**. Upon receiving the information indicating setting change is unnecessary, the mobile device **50** sends print data to the MFP **10** using the IP address included in the information indicating setting change is unnecessary and the wireless setting stored in the work area **58**. The MFP **10** receives the print data by using the wireless LAN I/F **20** (S**20**). Upon receiving the print data, the MFP **10** causes the print executing unit **16** to execute the print process.

(82) Moreover, in the sequence view of the present specification, the wireless communication executed by the MFP **10** by using the NFC I/F **22** (i.e., wireless communication according to the NFC system), and the wireless communication executed by the MFP **10** by using the wireless LAN I/F **20** (i.e., wireless communication according to the WFD system or normal Wi-Fi) is represented by arrows. The arrows representing the wireless communication using the wireless LAN I/F **20** are fatter than the arrows representing the wireless communication using the NFC I/F **22**.

(83) According to this configuration, in the case where the MFP **10** determines that the mobile device **50** currently belongs to the network to which the MFP **10** currently belongs, the MFP **10** can appropriately execute the communication of print data via the network to which the MFP **10** and the mobile device **50** currently belong without changing the wireless setting to which the MFP **10** and the mobile device **50** are currently set.

(84) (Second Situation)

(85) In the second situation shown in FIG. **4**, the MFP **10** currently belongs to the WFD network. The MFP **10** is operating in the G/O state in the WFD network. The PC **8** that is in the client state currently belongs to the WFD network. The mobile device **50** does not currently belong to the wireless network to which the MFP **10** currently belongs. The mobile device **50** may currently belong, or may not belong, to a wireless network other than the wireless network to which the MFP

10 currently belongs.
(86) In this situation, upon receiving the NFC information from the mobile device **50** by using the NFC I/F **22**, the MFP **10** determines in S**6** that the mobile device **50** does not currently belong to the WFD network to which the MFP **10** currently belongs (NO in S**6**). Moreover, in the case where the mobile device **50** currently belongs to a wireless network, the NFC information includes the SSID and BSSID of the wireless network. However, in the case where the mobile device **50** does not currently belong to a wireless network, the NFC information does not include the SSID and BSSID of a wireless network. In S**12**, the MFP **10** determines that the MFP **10** is in the G/O state (YES in S**12**). In this case, in S**16**, the MFP **10** sends the wireless setting of the MFP **10** stored in the work area **38** and the IP address of the MFP **10** to the mobile device **50** by using the NFC I/F **22**. Upon receiving the wireless setting, the mobile device **50** stores the received wireless setting in the work area **58**. Next, the MFP **10** and the mobile device **50** establish a WFD connection (S**18**). Thereby, the mobile device **50** can belong to the WFD network to which the MFP **10** currently belongs. Moreover, by using the NFC I/F **22**, the MFP **10** sends a wireless setting including the authentication method and the encryption method of the MFP **10** to the mobile device **50**. According to this configuration, the mobile device **50** can execute an authentication process according to the authentication method and encryption method received from the MFP **10**, and need not execute any process to verify whether an authentication method and encryption method is to be used. Consequently, the MFP **10** and the mobile device **50** can establish a connection comparatively promptly.
(87) Next, the mobile device **50** sends print data to the MFP **10** by using the wireless setting stored in the work area **58** and the IP address received in **516**. The MFP **10** receives the print data by using the wireless LAN I/F **20** (S**20**). Upon receiving the print data, the MFP **10** causes the print executing unit **16** to execute a print process. According to this configuration, in the case where the MFP **10** is operating in the G/O state in the WFD network, the MFP **10** can appropriately execute the communication of print data with the mobile device **50** via the WFD network to which the MFP **10** currently belongs.
(88) (Third Situation)
(89) In the third situation shown in FIG. **5**, the MFP **10** currently belongs to the WFD network. The MFP **10** is operating in the client state in the WFD network. The PC **8** that is in the G/O state currently belongs to the WFD network, whereas the mobile device **50** does not currently belong. The mobile device **50** is in the same state as in the second situation.
(90) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC I/F **22**, the MFP **10** determines in S**6** that the mobile device **50** does not currently belong to the WFD network to which the MFP **10** currently belongs (NO in S**6**). In S**10** the MFP **10** determines that the MFP **10** is in the client state (YES in S**10**). In this case, in S**14** the MFP **10** sends the communication NG information to the mobile device **50** by using the NFC I/F **22**.
(91) In this case, the MFP **10** does not send the wireless setting stored in the work area **38** to the mobile device **50**. According to this configuration, the wireless setting of the PC **8** that is operating in the G/O state in the WFD network does not need to be provided to the mobile device **50**. Thereby, it is possible to prevent the mobile device **50** from entering the WFD network. Further, by receiving the communication NG information from the MFP **10**, the mobile device **50** can notify the user of the mobile device **50** that the MFP **10** is not executing the communication of object data with the mobile device **50**.
(92) (Fourth Situation)
(93) In a fourth situation shown in FIG. **6**, the MFP **10** is set to the WFD=ON mode, but does not currently belong to a WFD network. That is, the MFP **10** is operating in the device state. Moreover, the state of the MFP **10** is either a state of currently belonging or not currently belonging to a normal Wi-Fi network. The mobile device **50** is in the same state as in the second situation.
(94) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC

I/F **22**, the MFP **10** determines in S**4** that the MFP **10** does not currently belong to a network (NO in S**4**: the case where the MFP **10** is in the state of not currently belonging to the normal Wi-Fi network), or the MFP **10** determines in S**6** that the mobile device **50** does not currently belong to the normal Wi-Fi network to which the MFP **10** currently belongs (YES in S**6**: the MFP **10** is in a state of currently belonging to a normal Wi-Fi network). Further, in S**10** and S**12**, the MFP **10** determines that the MFP **10** is not in either the G/O state or the client state (NO in both S**10**, S**12**). In this case, in S**15** the MFP **10** sets the MFP **10** to spontaneous G/O mode without executing the G/O negotiation.

(95) Next, in S**16**, the MFP **10** sends the wireless setting of the MFP **10** stored in the work area **38** (i.e., the wireless setting prepared at the stage of setting spontaneous G/O mode in S**15**) and the IP address of the MFP **10** to the mobile device **50** by using the NFC I/F **22**. Upon receiving the wireless setting, the mobile device **50** stores the received wireless setting in the work area **58**. Next, the MFP **10** and the mobile device **50** establish a WFD connection (S**18**). Thereby, the mobile device **50** can belong to the WFD network in which the MFP **10** is operating in the G/O state.

(96) Next, the mobile device **50** sends print data to the MFP **10** by using the wireless setting stored in the work area **58** and the IP address received in S**16**. The MFP **10** receives the print data by using the wireless LAN I/F **20** (S**20**). Upon receiving the print data, the MFP **10** causes the print executing unit **16** to execute the print process. According to this configuration, the MFP **10** can newly construct a WFD network in which the MFP **10** is operating in the G/O state in the WFD network. Thereby, the MFP **10** can appropriately execute the communication of print data with the mobile device **50** via the newly constructed WFD network. Further, since the MFP **10** is necessarily operating in the G/O state in the newly constructed WFD network, the MFP **10** can determine an authentication method, etc. to be used in the WFD network.

(97) (Fifth Situation)

(98) In the fifth situation shown in FIG. **7**, the MFP **10** is set to the WFD=OFF mode. Moreover, the state of the MFP **10** is either the state of currently belonging or not currently belonging to the normal Wi-Fi network. The mobile device **50** is in the same state as in the second situation.

(99) In this situation, upon receiving the NFC information from the mobile device **50** by using the NFC I/F **22**, NO is determined in S**4** or S**6** in the same manner as in the fourth situation. The MFP **10** determines in S**8** that the MFP **10** is set to the WFD=OFF mode. In this case, in S**9** the MFP **10** changes the mode from the WFD=OFF mode to the WFD=ON mode. Next, in S**15** the MFP **10** sets the MFP **10** to the spontaneous G/O mode.

(100) Below, the processes until the print process are the same as in the fourth situation. In this configuration, also, the same advantages as in the fourth situation can be achieved. When the print process ends, the MFP **10** determines that an external device does not currently belong to the newly constructed WFD network (NO in S**23**), and changes the mode from the WFD=ON mode to the WFD=OFF mode. According to this configuration, in the case where the external device does not belong to the WFD network after the communication of print data, the mode can appropriately be changed from the WFD=ON mode to the WFD=OFF mode.

(101) In the present embodiment, the MFP **10** can, by using the wireless LAN I/F **20**, appropriately execute the wireless communication of object data with the mobile device **50** at a comparatively fast communication speed by executing processes in accordance with whether the MFP **10** currently belongs to the same network as the mobile device **50**, i.e., in accordance with whether the MFP **10** is capable of communicating with the mobile device **50**. Further, the MFP **10** can execute the communication of object data with the mobile device **50** via the WFD network without the MFP **10** and the mobile device **50** communicating via different access points.

(102) Further, in the case where the MFP **10** is not capable of communicating with the mobile device **50** and the MFP **10** currently belongs to the WFD network, the MFP **10** can appropriately execute the communication of the object data with the mobile device **50** via the WFD network to which the MFP **10** currently belongs. Further, in the case where the MFP **10** does not currently

belong to the WFD network, the MFP **10** can appropriately execute the communication of the object data with the mobile device **50** via the newly constructed WFD network.
(103) (Corresponding Relationships)
(104) The MFP **10** is an example of the "communication device", the NFC I/F **22** is an example of the "first type of interface", and the wireless LAN I/F **20** is an example of the "second type of interface". Moreover, from the above description, since the NFC I/F **22** (i.e., the "first type of interface") executes communication using the wireless LAN I/F **20** (i.e., the "second type of interface"), the NFC I/F **22** can be called an interface used for communication executed between the MFP **10** (i.e., the "communication device") and the mobile device **50**.
(105) The AP **6** is an example of the "access point". That is, the "access point" is a device that, within a network to which the access point belongs, i.e., a normal Wi-Fi network, relays communication between a pair of apparatuses belonging to the normal Wi-Fi network.
(106) The NFC information is an example of the "specific information", and the SSID and BSSID included in the NFC information is the "wireless network identifier included in the specific information". The state of the mobile device **50** currently belonging to the network to which the MFP **10** currently belongs is the "communication-enabled state". The processes S**15** to S**18** are an example of the "specific process". Wireless communication via the WFD network by using the wireless LAN I/F **20** is an example of the "specific wireless communication". The G/O state is an example of the "parent station state", and the client state is an example of the "child station state". In the case where YES is determined in S**4**, the WFD network to which the MFP **10** belongs is an example of the "first wireless network", and the WFD network constructed by the processes S**15** to S**18** is an example of the "second wireless network". The communication NG information is an example of the "information indicating that the communication of object data is not executed". The WFD=ON mode is an example of the "first mode", and the WFD=OFF mode is an example of the "second mode".
Second Embodiment
(107) Points differing from the first embodiment will be described. In the present embodiment, a communication process of FIG. **8** is executed instead of the communication process of FIG. **2**. S**2** to S**24** of FIG. **8** are the same as the processes S**2** to S**24** of FIG. **2**. In the case of YES in S**10**, i.e., in the case where the MFP **10** currently belongs to the WFD network and is operating in the client state in the WFD network, in S**42** the determining unit **42** determines whether the MFP **10** is capable of disconnecting from the WFD network to which it currently belongs. Specifically, in the case of either the situation where data communication is currently being executed via the WFD network or the situation where data communication is to be executed via the WFD network, the determining unit **42** determines that the MFP **10** is not capable of disconnecting from the WFD network to which it currently belongs (NO in S**42**). On the other hand, in the case of neither the situation where data communication is being executed via the WFD network nor the situation where data communication is to be executed, the determining unit **42** determines that the MFP **10** is capable of disconnecting from the WFD network to which it currently belongs (YES in S**42**).
(108) For example, a case is assumed where the PC **8** is operating in the G/O state in a WFD network to which the MFP **10** currently belongs. In the case where the MFP **10** is currently receiving print data from the PC **8** by using the wireless LAN I/F **20**, the determining unit **42** determines a situation where data communication is currently being executed. Further, in the case where the MFP **10** is creating scan data in accordance with a scan instruction from the PC **8** and, once the scan data is created, the MFP **10** is to send the scan data to the PC **8** by using the wireless LAN I/F **20**, a situation where data communication is to be executed is determined.
(109) In the case of NO in S**42**, the process proceeds to S**14**, and in the case of YES in S**42**, the process proceeds to S**44**. In S**44**, the communication executing unit **44** disconnects the MFP **10** from the WFD network to which the MFP **10** is currently joined. Specifically, the communication executing unit **44** deletes the wireless setting that is being stored in the work area **38**, and changes

the state value in the work area **38** to a value representing the device state. Next, the communication executing unit **44** executes the process S**15**.

Advantages of Present Embodiment

(110) The MFP **10** of the second embodiment can achieve the same advantages as the MFP **10** of the first embodiment in the first, second, fourth, and fifth situations. The advantages of the present embodiment in sixth and seventh situations will be described with reference to FIGS. **9** and **10**. Moreover, processes corresponding to the communication process of FIG. **8** are shown in each of FIGS. **9** and **10**.

(111) (Sixth Situation)

(112) In the sixth situation shown in FIG. **9**, the MFP **10** currently belongs to a WFD network. The MFP **10** is operating in the client state in the WFD network. The PC **8** that is in the G/O state currently belongs to the WFD network, but the mobile device **50** does not currently belong to the WFD network. The mobile device **50** is in the same state as in the second situation. Further, the MFP **10** is receiving print data from the PC **8**.

(113) In this situation, when NFC information is received from the mobile device **50** by using the NFC I/F **22**, NO is determined in S**6** and YES is determined in S**10**, as in the third situation. Since the MFP **10** is receiving print data from the PC **8**, it is determined that the MFP **10** is not capable of disconnecting from the WFD network (NO in S**42**). In this case, in S**14** the MFP **10** sends the communication NG information to the mobile device **50** via the NFC I/F **22**.

(114) According to this configuration, in the case where the MFP **10** is executing data communication via the WFD network to which the MFP **10** currently belongs, or in the case where the MFP **10** is to execute data communication via the WFD network to which the MFP **10** currently belongs, the MFP **10** can be prevented from disconnecting from the WFD network to which it currently belongs.

(115) (Seventh Situation)

(116) In the seventh situation shown in FIG. **10**, the MFP **10** currently belongs to a WFD network. The MFP **10** is operating in the client state in the WFD network. The PC **8** that is in the G/O state currently belongs to the WFD network, but the mobile device **50** does not currently belong to the WFD network. However, the MFP **10** is not in the situation of currently executing the data communication with the PC **8**, nor is in the situation where it is to execute the data communication with the PC **8**. The mobile device **50** is in the same state as in the second situation.

(117) In this situation, when NFC information is received from the mobile device **50** by using the NFC I/F **22**, NO is determined in S**6** and YES is determined in S**10**, as in the sixth situation. Since the MFP **10** is not in the situation of executing the data communication via the WFD network nor in the situation where it is to execute the data communication, it is determined that the MFP **10** is capable of disconnecting from the WFD network to which it currently belongs (YES in S**42**). In this case, in S**15** the MFP **10** sets the MFP **10** to the spontaneous G/O mode. The subsequent processes are the same as the processes after the MFP **10** was set to the spontaneous G/O mode in the fourth situation.

(118) According to this configuration, the MFP **10** disconnects from the WFD network in which the MFP **10** is operating in the client state, and can newly construct a WFD network. Thereby, the MFP **10** can appropriately execute the communication of object data with the mobile device **50** via the newly constructed WFD network.

(119) (Corresponding Relationships)

(120) The processes S**42**, S**44** and the processes S**15** to S**18** of FIG. **8** are an example of the “specific process”.

Third Embodiment

(121) Points differing from the first embodiment will be described. In the present embodiment, a communication process of FIG. **11** is executed instead of the communication process of FIG. **2**. Moreover, in the present embodiment, the mobile device **50** sends, to the MFP **10**, NFC

information further including WFD-compatible information, indicating whether the mobile device **50** is capable of executing wireless communication according to the WFD system, and device ID of the mobile device **50** (e.g., MAC address, serial number, etc.).
(122) S**2** to S**24** of FIG. **11** are the same as the processes S**4** to **24** of FIG. **2**. In the case where the process S**9** is executed, and in the case where NO is determined in **512**, i.e., in the case where the MFP **10** is operating in the device state, the determining unit **42** determines in S**52**, by using the NFC information, whether the mobile device **50** is capable of executing wireless communication according to the WFD system. In the case where WFD-compatible information indicating that the mobile device **50** is capable of executing wireless communication according to the WFD system is included in the NFC information, the determining unit **42** determines that the mobile device **50** is capable of executing wireless communication according to the WFD system (YES in S**52**), and the process proceeds to S**54**.
(123) On the other hand, in the case where WFD-compatible information indicating that the mobile device **50** is capable of executing wireless communication according to the WFD system is not included in the NFC information, the determining unit **42** determines that the mobile device **50** is not capable of executing wireless communication according to the WFD system (NO in S**52**), and the process proceeds to S**15**.
(124) In S**54** the communication executing unit **44** sends WFD connection start information indicating that the WFD connection is being started to the mobile device **50** via the NFC I/F **22**. A WPS (abbreviation of: Wi-Fi Protected Setup) wireless connection system is used as a system for executing the WFD system wireless connection. WPS wireless connection systems include a PBC (abbreviation of: Push Button Configuration) system and a PIN (abbreviation of: Personal Identification Number) code system. In the present embodiment, the PBC code system will be described. However, the technique of the present embodiment can also be applied to the PIN code system. The WFD connection start information includes information indicating that the PBC code system is used as the system for executing the WFD system wireless connection. The WFD connection start information further includes a device ID of the MFP **10** (e.g., MAC address, serial number, etc.). Thereby, the mobile device **50** that receives the WFD connection start information can recognize an apparatus (i.e., the MFP **10**) identified by the device ID included in the WFD connection start information, and that processes S**58**, S**62** (to be described) are to be executed.
(125) Upon receiving the WFD connection start information, the mobile device **50** determines whether the mobile device **50** is set so as to be capable of executing the wireless communication according to the WFD system. In the case of being set so as to be capable of executing the wireless communication according to the WFD system, the mobile device **50** maintains the wireless LAN I/F setting, and in the case of not being set so as to be capable of executing the wireless communication according to the WFD system, the mobile device **50** changes to a setting allowing it to be capable of executing the wireless communication according to the WFD system.
(126) Next, in S**55** the communication executing unit **44** searches for the mobile device **50**. Specifically, the communication executing unit **44** sequentially executes a Scan process, a Listen process, and a Search process. The Scan process is a process for searching for a G/O state apparatus present in the surroundings of the MFP **10**. Specifically, in the Scan process, the communication executing unit **44** wirelessly sends a Probe Request signal, sequentially, by using 13 channels 1 ch to 13 ch sequentially. Moreover, this Probe Request signal includes P2P (Peer 2 Peer) information indicating that the MFP **10** is capable of executing the WFD function.
(127) For example, in the case where a G/O state WFD-compatible apparatus (called “specific G/O apparatus” below) is present in the surroundings of the MFP **10**, it is predetermined that the specific G/O apparatus uses one channel from among 1 ch to 13 ch. Consequently, the specific G/O apparatus wirelessly receives a Probe Request signal from the MFP **10**. In this case, the specific G/O apparatus wirelessly sends a Probe Response signal to the MFP **10**. This Probe Response signal includes P2P information indicating that the specific G/O apparatus is capable of executing

the WFD function, and information indicating that the specific G/O apparatus is in the G/O state. Consequently, the communication executing unit **44** can find the specific G/O apparatus. Moreover, the Probe Response signal further includes information indicating a device name of the specific G/O apparatus and a category (e.g., mobile device, PC, etc.) of the specific G/O apparatus, and a MAC address of the specific G/O apparatus. Consequently, the communication executing unit **44** can acquire information related to the specific G/O apparatus.
(128) In the case where the device ID of the specific G/O apparatus included in the Probe Response signal and the device ID of the mobile device **50** included in the NFC information are identical, the communication executing unit **44** can identify that the specific G/O apparatus is the mobile device **50**. That is, in the case where the mobile device **50** currently belongs to the WFD network and the mobile device **50** is operating in the G/O state in the WFD network, the communication executing unit **44** can find the mobile device **50** by means of the Scan process.
(129) Moreover, for example, in the case where a device state WFD-compatible apparatus (called "specific device apparatus" below) is present in the surroundings of the MFP **10**, it is predetermined that the specific device apparatus uses one channel from among 1 ch, 6 ch, and 11 ch. Consequently, the specific device apparatus also wirelessly receives a Probe Request signal from the MFP **10**. In this case, the specific device apparatus wirelessly sends a Probe Response signal to the MFP **10**. However, this Probe Response signal includes information indicating that the specific device apparatus is in the device state, and does not include information indicating that the specific device apparatus is in the G/O state. Further, even if an apparatus that is in the client state wirelessly receives the Probe Request signal from the MFP **10**, the client state apparatus does not wirelessly send the Probe Response signal to the MFP **10**. Consequently, in the Scan process, the communication executing unit **44** can find the mobile device **50** in either the G/O state or the device state.
(130) The Listen process is a process for responding to the Probe Request signal. The specific device apparatus can wirelessly send the Probe Request signal during the Search process (to be described). That is, in the case where the current state of the mobile device **50** is the device state, the mobile device **50** periodically sends the Probe Request signal wirelessly. This Probe Request signal includes the device ID of the mobile device **50** (e.g., MAC address, serial number, etc.).
(131) In the case where the device ID of the specific device apparatus included in the Probe Request signal and the device ID of the mobile device **50** included in the NFC information are identical, the communication executing unit **44** can identify that the specific device apparatus is the mobile device **50**. That is, in the case where the mobile device **50** is operating in the device state, the communication executing unit **44** can find the mobile device **50** by means of the Listen process. Upon receiving the Probe Request signal from the mobile device **50**, the communication executing unit **44** wirelessly sends a Probe Response signal.
(132) In the Search process, the communication executing unit **44** sequentially uses the three channels 1 ch, 6 ch, 11 ch to sequentially send the Probe Request signal wirelessly. Thereby, the communication executing unit **44** wirelessly receives the Probe Response signal from the specific device apparatus. This Probe Response signal includes the P2P information indicating that the specific device apparatus is capable of executing the WFD function, information indicating that the specific device apparatus is in the device state, and the device ID of the specific device apparatus (e.g., MAC address, serial number, etc.). In the case where the current state of the mobile device **50** is the device state, the mobile device **50** wirelessly sends the Probe Response signal in response to the Probe Request signal sent from the MFP **10**.
(133) In the case where the device ID of the specific device apparatus included in the Probe Response signal and the device ID of the mobile device **50** included in the NFC information are identical, the communication executing unit **44** can identify that the specific device apparatus is the mobile device **50**. That is, in the case where the mobile device **50** currently belongs to the WFD network and is operating in the device state in the WFD network, the communication executing

unit **44** can find the mobile device **50** by means of the Search process.
(134) In S**55** the communication executing unit **44** can find the mobile device **50** (YES in S**56**) both in the case where the mobile device **50** is operating in the G/O state and in the case where the mobile device **50** is operating in the device state. In the case where the mobile device **50** was not found in S**56** (NO in S**56**), the process proceeds to S**14**.
(135) In the case where the mobile device **50** is found (YES in S**56**), in S**57** the communication executing unit **44** determines whether the found mobile device **50** is in the device state. Specifically, in the case where information indicating that the mobile device **50** is in the device state is received in the process S**55**, it is determined that the mobile device **50** is in the device state (YES in S**57**), and the process proceeds to S**58**. On the other hand, in the case where information indicating that the mobile device **50** is in the device state is not received in the process S**55**, it is determined that the mobile device **50** is not in the device state (i.e., the mobile device **50** is in the G/O state) (NO in S**57**), and the process proceeds to S**62**.
(136) In S**58**, by using the wireless LAN I/F **20**, the communication executing unit **44** executes the G/O negotiation with the mobile device **50**, determining that one apparatus of the MFP **10** and the mobile device **50** is to operate in the G/O state and the other apparatus is to operate in the client state.
(137) Specifically, the communication executing unit **44** first wirelessly sends a connection request signal to the mobile device **50**. Consequently, the mobile device **50** also wirelessly sends an OK signal to the MFP **10**. Next, the communication executing unit **44** wirelessly sends information indicating G/O priority of the MFP **10** to the mobile device **50**, and receives information indicating G/O priority of the mobile device **50** from the mobile device **50**. Moreover, the G/O priority of the MFP **10** is an index indicating the degree to which the MFP **10** should become the G/O, and is predetermined in the MFP **10**. Similarly, the G/O priority of the mobile device **50** is an index indicating the degree to which the mobile device **50** should become the G/O. For example, an apparatus in which the capacity of the CPU and the memory is comparatively high (e.g. the MFP **10**) can execute another process rapidly while operating as the G/O. Consequently, the G/O priority is usually set in this type of apparatus so that it has a high possibility of becoming the G/O. On the other hand, e.g., an apparatus in which the capacity of the CPU and the memory is comparatively low (e.g., the mobile device **50**) might be unable to execute another process rapidly while operating as the G/O. Consequently, the G/O priority is usually set in this type of apparatus so that it has a low possibility of becoming the G/O.
(138) The communication executing unit **44** compares the G/O priority of the MFP **10** and the G/O priority of the mobile device **50**, and determines that the apparatus with high priority (the MFP **10** or the mobile device **50**) is to operate in the G/O state, and the apparatus with low priority (the MFP **10** or the mobile device **50**) is to operate in the client state. In the case of determining that the MFP **10** is to operate in the G/O state, the communication executing unit **44** changes the state value in the memory **34** from the value corresponding to the device state to the value corresponding to the G/O state. Consequently, the MFP **10** becomes able to operate in the G/O state. Further, in the case of determining that the MFP **10** is to operate in the client state, the communication executing unit **44** changes the state value in the memory **34** from the value corresponding to the device state to the value corresponding to the client state. Consequently, the MFP **10** becomes able to operate in the client state. Moreover, the G/O state and the client state of the mobile device **50** are determined based on the G/O priority of the MFP **10** and the G/O priority of a target apparatus by using the same method as the MFP **10**. When the G/O negotiation of S**58** ends, the process proceeds to S**62**.
(139) In S**62** the communication executing unit **44** establishes a connection between the MFP **10** and the mobile device **50** according to WPS. Specifically, the communication executing unit **44** determines whether the current state of the MFP **10** is the G/O state and the current state of the mobile device **50** is the client state or not. In the case where the current state of the MFP **10** is the G/O state and the current state of the mobile device **50** is the client state, the communication

executing unit **44** executes WPS negotiation for the G/O state.
(140) Specifically, the communication executing unit **44** creates the wireless setting needed to establish the wireless connection (SSID, authentication method, encryption method, password, etc.), and wirelessly sends it to the mobile device **50**. Moreover, the authentication method and encryption method are predetermined. Further, the communication executing unit **44** creates a password at the time of creating the wireless setting. Moreover, the SSID may be created by the communication executing unit **44**, or may be predetermined. Sending the wireless setting to the mobile device **50** allows the MFP **10** and the mobile device **50** to use the same wireless setting. That is, by using the wireless setting, the MFP **10** and the mobile device **50** execute the wireless communication of an Authentication Request, Authentication Response, Association Request, Association Response, and 4-way handshake. Various authentication processes such as authentication of the SSID, authentication of the authentication method and encryption method, authentication of the password, etc. are executed during this process. In a case where all the authentications succeed, a wireless connection is established between the MFP **10** and the mobile device **50**. Thereby, the state is achieved where the MFP **10** and the mobile device **50** belong to the same WFD network.
(141) On the other hand, in the case where the current state of the MFP **10** is the client state and the current state of the target apparatus is the G/O state, the communication executing unit **44** executes the WPS negotiation for the client state. Specifically, the mobile device **50** creates the wireless setting needed to establish the wireless connection (SSID, authentication method, encryption method, password, etc.), and wirelessly sends it to the MFP **10**. Consequently, the communication executing unit **44** wirelessly receives the wireless setting from the mobile device **50**. The subsequent processes (the communication processes of the Authentication Request, etc.) are the same as in the WPS negotiation for the G/O state. Thereby, a state is achieved where the MFP **10** and the mobile device **50** belong to the same WFD network. Consequently, it becomes possible to execute the wireless communication of object data (print data, etc.) between the MFP **10** that is in the client state and the mobile device **50** that is in the G/O state. When S**62** ends, the control unit **30** executes the processes S**20** to S**24** of FIG. **2**, ending the communication process.
Advantages of Present Embodiment
(142) The MFP **10** of the third embodiment can achieve the same advantages as the MFP **10** of the first embodiment in the first to third situations. The advantages of the present embodiment in eighth and ninth situations will be described with reference to FIGS. **12**, **13**. Moreover, processes corresponding to the communication process of FIG. **11** are shown in each of FIGS. **12**, **13**.
(143) (Eighth Situation)
(144) In the eighth situation shown in FIG. **12**, the MFP **10** is set to the WFD=ON mode, but does not currently belong to the WFD network. That is, the MFP **10** is operating in the device state. Moreover, the state of the MFP **10** is either the state of currently belonging or not currently belonging to the normal Wi-Fi network. The mobile device **50** does not currently belong to a wireless network.
(145) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC I/F **22**, the MFP **10** determines NO in S**4** or S**6**, and determines NO in S**10** and S**12**, as in the fourth situation.
(146) The MFP **10** sends the WFD connection start information to the mobile device **50** by using the wireless LAN I/F **20** (S**54**). Next, the MFP **10** executes the Search process of S**55**, searching for the mobile device **50**. Upon finding the mobile device **50** (YES in S**56**), the MFP **10** determines whether the found mobile device **50** is in the device state (S**57**). In the case of determining that the mobile device **50** is in the device state (YES in S**57**), the G/O negotiation (S**58**) and the WPS negotiation (S**62**) are executed by using the wireless LAN I/F **20**. Thereby, the WFD network is constructed to which the MFP **10** and the mobile device **50** belong.
(147) Next, the mobile device **50** sends print data to the MFP **10**. The MFP **10** receives the print

data by using the wireless LAN I/F **20** (S**20**). Upon receiving the print data, the MFP **10** causes the print executing unit **16** to execute the print process.
(148) (Ninth Situation)
(149) In the ninth situation of FIG. **13**, the MFP **10** is set to the WFD=OFF mode. Moreover, the state of the MFP **10** is either a state of currently belonging or not currently belonging to the normal Wi-Fi network. The mobile device **50** does not currently belong to the wireless network.
(150) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC I/F **22**, the MFP **10** determines NO in S**4** or S**6**, and determines NO in S**8**, as in the fifth situation. In this case, in S**9** the MFP **10** changes from the WFD=OFF mode to the WFD=ON mode.
(151) After changing from the WFD=OFF mode to the WFD=ON mode, the processes executed until the print process are the same as the processes executed until the print process in the eighth situation after NO was determined in S**10** and S**12**. When the print process ends, the MFP **10** determines that an external device does not currently belong to the newly constructed WFD network (NO in S**23**), and changes from the WFD=ON mode to the WFD=OFF mode. According to this configuration, in the case where the external device does not belong to the WFD network after the communication of print data, the mode can change appropriately from the WFD=ON mode to the WFD=OFF mode.
(152) According to this configuration, the MFP **10** can construct, with the mobile device **50**, the WFD network operating in either the G/O state or the client state. Thereby, the MFP **10** can appropriately execute the communication of the print data with the mobile device **50**.
(153) (Corresponding Relationships)
(154) The processes S**15** to S**18** and the processes S**52** to S**62** of FIG. **11** are an example of the “specific process”.
Fourth Embodiment
(155) Points differing from the first embodiment will be described. In the present embodiment, a communication process of FIG. **14** is executed instead of the communication process of FIG. **2**. S**2** to S**24** of FIG. **14** are the same as the processes S**2** to S**24** of FIG. **2**. In the case of NO in S**8**, i.e., in the case where the MFP **10** is not set to the WFD=ON mode, in S**76** the determining unit **42** determines whether the MFP **10** currently belongs to the normal Wi-Fi network. In the case where information indicating that the MFP **10** currently belongs to the normal Wi-Fi network is being stored in the work area **38**, the determining unit **42** determines that the MFP **10** currently belongs to the normal Wi-Fi network (YES in S**76**), and the process proceeds to S**80**. On the other hand, in the case where information indicating that the MFP **10** currently belongs to the normal Wi-Fi network is not being stored in the work area **38**, the determining unit **42** determines that the MFP **10** does not currently belong to the normal Wi-Fi network (NO in S**76**), and the process proceeds to S**9**.
(156) In the case of YES in S**10**, i.e., in the case where the MFP **10** currently belongs to a WFD network and is operating in the client state in the WFD network, in S**72** the communication executing unit **44** sends the G/O wireless setting without including the password, that is being stored in the work area **38** to the mobile device **50** via the NFC I/F **22**, and the process proceeds to S**20**.
(157) Upon receiving the G/O wireless setting, the mobile device **50** causes the user to specify a password. When a password has been specified by the user, the mobile device **50** establishes the connection with the G/O state device by using the wireless setting received from the MFP **10** and the password specified by the user. Thereby, the mobile device **50** becomes capable of wireless communication with the MFP **10** via the G/O state device. Moreover, in the case where the connection cannot be established between the mobile device **50** and the G/O state device, the MFP **10** cannot execute wireless communication with the mobile device **50**. In this case, the control unit **30** returns to S**2** without executing the processes S**20** to S**24**.
(158) In the case of YES in **512**, i.e., in the case where the MFP **10** currently belongs to a WFD

network and is operating in the G/O state in the WFD network, the process proceeds to S**13**. In S**16** the communication executing unit **44** sends the wireless setting of the MFP **10** stored in the work area **38** to the mobile device **50** via the NFC I/F **22**, and the process proceeds to S**18**. The wireless setting of the MFP **10** sent in S**16** includes the password.
(159) Further, in the case where NO is determined in **512**, i.e., in the case where the MFP **10** is operating in the device state, the determining unit **42** executes the process S**78**. The process S**78** is the same as the process S**76**. In the case of NO in S**78**, the process proceeds to S**15**, and in the case of YES in S**78**, the process proceeds to S**80**.
(160) In S**80**, the communication executing unit **44** sends the wireless setting not including the password, for belonging to the normal Wi-Fi network that is being stored in the work area **38**, i.e., the wireless setting of the AP (e.g., the AP **6**) to the mobile device **50** via the NFC I/F **22**, and the process proceeds to S**20**. Upon receiving the AP wireless setting, as in the case of S**72**, the mobile device **50** establishes a connection with the AP by using the wireless setting received from the MFP **10** and the password specified by the user. Thereby, the mobile device **50** becomes capable of wireless communication with the MFP **10** via the AP. Moreover, in the case where a connection cannot be established between the mobile device **50** and the AP, the MFP **10** cannot execute wireless communication with the mobile device **50**. In this case, the control unit **30** returns to S**2** without executing the processes S**20** to S**24**.
Advantages of Present Embodiment
(161) The MFP **10** of the fourth embodiment can achieve the same advantages as the MFP **10** of the first embodiment in the first, second, fourth and fifth situations. The advantages of the present embodiment in tenth and eleventh situations will be described with reference to FIGS. **15**, **16**. Moreover, processes corresponding to the communication process of FIG. **14** are shown in each of FIGS. **15**, **16**.
(162) (Tenth Situation)
(163) In the tenth situation shown in FIG. **15**, the MFP **10** currently belongs to the WFD network. The MFP **10** is operating in the client state in the WFD network. The PC **8** that is in the G/O state currently belongs to the WFD network, but the mobile device **50** does not currently belong to the WFD network. The mobile device **50** is in the same state as in the second situation.
(164) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC I/F **22**, the MFP **10** determines NO in S**6** and determines YES in S**10**, as in the third situation. In S**72**, the MFP **10** sends the IP address of the MFP **10** and the wireless setting not including the password, of the PC **8** that is operating in the G/O state to the mobile device **50** via the NFC I/F **22**. According to this configuration, a password does not need to be provided to the mobile device **50** and the user. Consequently, in the case where the mobile device **50** and the user do not know the password for belonging to the WFD network in which the PC **8** is operating in the G/O state, it is possible to prevent the mobile device **50** from entering the WFD network.
(165) Upon receiving the wireless setting, the mobile device **50** stores the received wireless setting in the work area **58**. Next, the mobile device **50** displays a password specifying screen on a displaying unit of the mobile device **50**. The user can specify the password by operating the control unit of the mobile device **50**. When the password is specified by the user, the mobile device **50** establishes the WFD connection with the PC **8**. Thereby, the mobile device **50** can belong to the WFD network to which the MFP **10** currently belongs. The mobile device **50** operates in the client state in the WFD network.
(166) Moreover, in a modification, the wireless setting used previously to belong to the network may be stored by the mobile device **50** in the memory of the mobile device **50**. In this case, upon receiving the wireless setting that does not include the password, the mobile device **50** may identify, from the memory of the mobile device **50**, the wireless setting that includes the same SSID as the SSID included in the received wireless setting. The mobile device **50** may establish the WFD connection with the PC **8** by using the wireless setting identified from the memory of the mobile

device **50**.
(167) Upon belonging to the WFD network, the mobile device **50** sends print data to the MFP **10** by executing wireless communication via the PC **8** by using the wireless setting stored in the work area **58** and the IP address of the MFP **10** received in S**72**. The MFP **10** receives the print data from the PC **8** by using the wireless LAN I/F **20** (S**20**) and, upon receiving the print data, causes the print executing unit **16** to execute the print process.
(168) According to this configuration, in the case where the MFP **10** does not belong to the same network as the mobile device **50**, the MFP **10** can appropriately execute the communication of print data with the mobile device **50** via the WFD network to which the MFP **10** currently belongs.
(169) (Eleventh Situation)
(170) In the eleventh situation shown in FIG. **16**, the MFP **10** currently belongs to a normal Wi-Fi network. The MFP **10** is connected with the AP **6** in the normal Wi-Fi network. The mobile device **50** is in the same state as in the second situation.
(171) In this situation, upon receiving NFC information from the mobile device **50** by using the NFC I/F **22**, in S**6** the MFP **10** determines that the mobile device **50** does not currently belong to the normal Wi-Fi network to which the MFP **10** currently belongs (NO in S**6**). Further, in S**8** the MFP **10** determines that it is set to WFD=OFF mode (NO in S**8**). Next, in S**76** the MFP **10** determines that it currently belongs to the normal Wi-Fi network (YES in S**76**). In this case, in S**80** the MFP **10** sends, to the mobile device **50**, the IP address of the MFP **10** and the wireless setting of the AP **6** not including the password, which is stored in the work area **38**. According to this configuration, the password does not need to be provided to the mobile device **50** and the user. Consequently, in the case where the mobile device **50** and the user do not know the password for belonging to the normal Wi-Fi network in which the AP **6** is used, it is possible to prevent the mobile device **50** from entering the normal Wi-Fi network.
(172) When the wireless setting is received, as in the tenth situation, the mobile device **50** receives the wireless setting and causes the user to specify the password. Next, when the password is specified by the user, the mobile device **50** establishes a normal Wi-Fi connection with the AP **6**. Thereby, the mobile device **50** can belong to the normal Wi-Fi network to which the MFP **10** currently belongs. The mobile device **50** can send the print data to the MFP **10** via the AP **6**. Upon belonging to the normal Wi-Fi network, the mobile device **50** executes the wireless communication via the AP **6** by using the wireless setting stored in the work area **58** and the IP address of the MFP **10** received in S**80**, thereby sending the print data to the MFP **10**.
(173) Moreover, although not shown, in the case where the MFP **10** is currently operating in the device state (YES in S**8**, NO in S**10** and S**12**) and the MFP **10** belongs to the normal Wi-Fi network (YES in S**78**), as well, the MFP **10** sends, to the mobile device **50**, the AP wireless setting not including the password, that is being stored in the work area **38**.
(174) According to this configuration, in the case where the MFP **10** does not belong to the same network as the mobile device **50**, the MFP **10** can appropriately execute the communication of the print data with the mobile device **50** via the normal Wi-Fi network to which the MFP **10** currently belongs.
(175) (Corresponding Relationships)
(176) The processes S**15** to S**18**, the process S**72**, the processes S**7** to S**18**, and the process S**80** of FIG. **14** are an example of the “specific process”. In the case where YES is determined in S**4** of FIG. **14**, the WFD network and the normal Wi-Fi network to which the MFP **10** belongs are an example of the “first wireless network”.
Fifth Embodiment
(177) Points differing from the first embodiment will be described. In the present embodiment, in the case where the mobile device **50** currently belongs to a network, the mobile device **50** sends, to the MFP **10**, NFC information further including the password, authentication method and encryption method as the wireless setting stored in the work area **58**.

(178) Further, in the present embodiment, a communication process of FIG. **17** is executed instead of the communication process of FIG. **2**. S**2** to S**24** of FIG. **17** are the same as the processes S**2** to S**24** of FIG. **2**. In the case of NO in S**4** (i.e., in the case where the MFP **10** does not currently belong to a network), or in the case of NO in S**6** (i.e., in the case where the MFP **10** and the mobile device **50** do not belong to the same network), in S**82** the determining unit **42** determines whether the wireless setting is included in the NFC information received from the mobile device **50** by using the NFC I/F **22**. In the case where it is determined that the wireless setting is included (YES in S**82**), in S**83** the information indicating setting change is unnecessary is sent to the mobile device **50** by using the NFC I/F **22**. Next, in S**84**, by using the wireless setting included in the NFC information, the communication executing unit **44** joins the network to which the mobile device **50** belongs, and the process proceeds to S**20**.
(179) On the other hand, in the case where it is determined that the wireless setting is not included in the NFC information (NO in S**82**), the process proceeds to S**8**.
Advantages of Present Embodiment
(180) The MFP **10** of the fifth embodiment can achieve the same advantages as the MFP **10** of the first embodiment in the first to fifth situations. The advantages of the present embodiment in a twelfth situation will be described with reference to FIG. **18**. Moreover, processes corresponding to the communication process of FIG. **17** are shown in FIG. **18**.
(181) (Twelfth Situation)
(182) In the twelfth situation shown in FIG. **18**, the MFP **10** does not currently belong to a network, or currently belongs to a network to which the mobile device **50** does not belong. On the other hand, the mobile device **50** currently belongs to a normal Wi-Fi network to which the AP **6** belongs.
(183) In this situation, upon receiving NFC information from the mobile device **50** via the NFC I/F **22**, the MFP **10** determines that the MFP **10** and the mobile device **50** do not currently belong to the same network (NO in S**4** or NO in S**6**). Next, in S**82** the MFP **10** determines that the wireless setting for belonging to the network to which the mobile device **50** currently belongs is included in the NFC information (YES in S**82**).
(184) Next, in S**83** the MFP **10** sends the information indicating setting change is unnecessary to the mobile device **50** by using the NFC I/F **22**. Moreover, in the case of NO in S**4** (i.e., the case where the MFP **10** does not currently belong to a network), the MFP **10** sends information indicating setting change is unnecessary including the MAC address of the MFP **10** to the mobile device **50**. Further, in the case of NO in S**6** (i.e., the case where the MFP **10** currently belongs to a network, but the MFP **10** and the mobile device **50** do not belong to the same network), the MFP **10** sends information indicating setting change is unnecessary including the IP address of the MFP **10** to the mobile device **50**.
(185) The MFP **10** establishes a normal Wi-Fi connection with the AP **6** by using the wireless setting included in the NFC information (S**84**). In the case where the IP address of the MFP **10** is included in the information indicating setting change is unnecessary, the mobile device **50** specifies that IP address in the destination, and sends print data to the MFP **10** via the AP **6** (S**20**). Further, in the case where the MAC address of the MFP **10** is included in the information indicating setting change is unnecessary, the mobile device **50** identifies the IP address of the MFP **10** in accordance with RARP (abbreviation of. Reverse Address Resolution Protocol), specifies the identified IP address in the destination, and sends print data to the MFP **10** via the AP **6** (S**20**).
(186) According to this configuration, in the case where the MFP **10** and the mobile device **50** do not belong to the same network, the MFP **10** can appropriately execute the communication of print data with the mobile device **50** via the network to which the mobile device **50** currently belongs.
(187) (Corresponding Relationships)
(188) The processes S**15** to S**18** and the processes S**83**, S**84** of FIG. **17** are an example of the "specific process". In the case where YES is determined in S**82** of FIG. **17**, the network to which the mobile device **50** belongs is an example of the "third wireless network".

Sixth Embodiment

(189) Points differing from the fourth embodiment will be described. In the present embodiment, a communication process of FIG. **19** is executed instead of the communication process of FIG. **14**. In the communication process of FIG. **19**, in the case where YES is determined in S**10**, a process the same as S**14** of FIG. **2** is executed without the process S**72** of FIG. **14** being executed.

(190) According to this configuration, the same advantages as in the third situation can be achieved.

Seventh Embodiment

(191) Points differing from the first embodiment will be described. The MFP **10** of the present embodiment comprises a wired LAN I/F (not shown) in addition to the I/Fs **20**, **22**. In the present embodiment, a communication process of FIG. **20** is executed instead of the communication process of FIG. **2**. S**2**, S**8** to S**24** of FIG. **20** are the same as the processes of S**2**, S**8** to S**24** of FIG. **2**.

(192) As shown in FIG. **20**, when NFC information is received by using the NFC I/F **22** (YES in S**2**), in S**94** the determining unit **42** determines whether the MFP **10** currently belongs to a wired network that uses the wired LAN I/F and, further, whether the MFP **10** currently belongs to the wireless network that uses the wireless LAN I/F **20**. Specifically, in the case where the MFP **10** currently belongs to the wired network by using the wired LAN I/F, information indicating that the MFP **10** currently belongs to the wired network is stored in the work area **38** of the memory **34**. In the case where the information indicating that the MFP **10** currently belongs to the wired network is being stored in the work area **38**, the determining unit **42** determines that the MFP **10** currently belongs to a wired network (YES in S**94**). Further, in the case where the information indicating that the MFP **10** currently belongs to the wired network is not being stored in the work area **38**, the determining unit **42** determines that the MFP **10** does not currently belong to the wired network (NO in S**94**). The determination as to whether the MFP **10** currently belongs to the wireless network is the same as in the first embodiment.

(193) If it is determined in S**94** that the MFP **10** currently belongs to the wired network or the wireless network, or to both networks (YES in S**94**), the process proceeds to S**96**. If it is determined in S**94** that the MFP **10** does not belong to either the wired network or the wireless network (NO in S**94**), the process proceeds to S**8**.

(194) In S**96** the control unit **30** determines whether the MFP **10** is in a state of being capable of executing the communication of the object data with the mobile device **50** via the network to which the MFP **10** currently belongs (either the wired network or the wireless network). Specifically, this will be described according to a thirteenth situation of FIG. **21**.

(195) FIG. **21** shows an example in which the MFP **10** is connected with the AP **6** via the wired LAN, and the mobile device **50** is connected with the AP **6** via the normal Wi-Fi network.

(196) Moreover, in the sequence view of FIG. **21**, communication using the wired LAN I/F is represented by arrows. The arrows representing communication using the wired LAN I/F are fatter than the arrows representing wireless communication using the wireless LAN I/F **20**.

(197) The MFP **10** receives NFC information from the mobile device **50**. The NFC information sent from the mobile device **50** includes the device ID and the IP address of the mobile device **50**. In S**96** the determining unit **42** determines whether the MFP **10** is capable of communicating with the mobile device **50** via the network to which the MFP **10** currently belongs (the wired LAN in the example of FIG. **21**). Specifically, the determining unit **42** unicasts a device ID inquiry by using the wired LAN I/F. The determining unit **42** specifies the IP address included in the NFC information as the destination of the inquiry, and sends the inquiry.

(198) Upon receiving the inquiry via the AP **6**, the mobile device **50** sends the device ID of the mobile device **50** to the MFP **10** that is the source of the inquiry. Upon receiving the device ID of the mobile device **50** via the AP **6**, by using the wired LAN I/F, the determining unit **42** determines whether the received device ID and the device ID included in the NFC information are identical. In

the case where the two device IDs are identical, the determining unit **42** determines YES in S**96**, and proceeds to S**7**. On the other hand, in the case where there is no response to the inquiry or the two device IDs are not identical, the determining unit **42** determines NO in S**96**, and proceeds to S**8**.
(199) According to this configuration, in the case where the current state of the MFP **10** is a state capable of communicating with the mobile device **50**, the MFP **10** can execute the communication of object data with the mobile device **50** by using the wired LAN I/F.
(200) In the first embodiment, it is determined in S**6** whether the MFP **10** and the mobile device **50** are present in the same network. By contrast, in the present embodiment, in S**96** it is determined whether the MFP **10** and the mobile device **50** are capable of communication regardless of whether the MFP **10** and the mobile device **50** are present in the same network.
(201) (Corresponding Relationships)
(202) The wireless LAN I/F and wired LAN I/F are an example of the "second type of interface". The wired LAN is an example of the "first wireless network". The device ID included in the NFC information is an example of the "terminal identification information".
Modifications
(203) (1) The "communication device" is not restricted to the multi-function peripheral, but may be another apparatus comprising the first type of interface and the second type of interface (e.g., printer, FAX device, copier, scanner, etc.).
(204) (2) The MFP **10** may store an AP program for functioning as an access point. Upon activation of the AP program, the control unit **30** may store a predetermined wireless setting in the work area **38**. For example, in S**15** of FIG. **2**, the communication executing unit **44** may activate the AP program instead of setting the MFP **10** to spontaneous G/O mode. Next, the communication executing unit **44** may send the wireless setting that has been pre-stored in the work area **38** to the mobile device **50**. Thereupon, the communication executing unit **44** and the mobile device **50** may establish a connection by using the wireless setting pre-stored in the work area **38**. In this case, the MFP **10** may establish a normal Wi-Fi connection with the mobile device **50** and, further, the MFP **10** may construct a normal Wi-Fi network. In the present modification, the normal Wi-Fi which the MFP **10** executes with the mobile device **50** by activating the AP program and functioning as the access point is an example of the "specific wireless communication". Further, the activation of the AP program, sending of the wireless setting, and establishment of the normal Wi-Fi connection is an example of the "specific process".
(205) (3) The combination of the "first type of interface" and the "second type of interface" is not restricted to the combination of the NFC I/F and the wireless LAN I/F. For example, in the case where the wireless LAN I/F is adopted as the "second type of interface", the "first type of interface" may be an interface for executing infrared communication, an interface for executing Bluetooth (registered trademark), or an interface for executing Transfer Jet. Further, in the case where the NFC I/F is adopted as the "first type of interface", the "second type of interface" may be an interface for executing wired communication, or an interface for executing Bluetooth (registered trademark). In general terms, the combination of the interfaces may be any combination whereby the communication speed of communication via the second type of interface is faster than the communication speed of communication via the first type of interface.
(206) (4) The "first type of interface" and the "second type of interface" may physically be two interfaces (i.e., two separate IC chips), as in the above embodiments, or may physically be one interface (i.e., two types of communication are realized with one IC chip).
(207) (5) In the above embodiments, the interface for executing wireless communication according to the WFD system and the interface for executing wireless communication according to normal Wi-Fi was physically one interface (the wireless LAN I/F **20**). However, it may physically be a plurality of interfaces (i.e., two separate IC chips). In the present modification, the plurality of interfaces is an example of the "second type of interface".

(208) (6) In the first, second, and fourth to seventh embodiments, in S**15** the communication executing unit **44** sets the MFP **10** to spontaneous G/O mode. However, in the case where the mobile device **50** is capable of executing wireless communication according to the WFD system, the communication executing unit **44** may execute the processes S**54** to S**62** of FIG. **11** instead of S**15** to S**18**. In the present modification, the processes S**54** to S**62** are an example of the “specific process”.
(209) (7) In the seventh embodiment, the determining unit **42** specifies the IP address included in the NFC information (i.e., the IP address of the mobile device **50**) in the destination, and sends the device ID inquiry. However, the determining unit **42** may broadcast the device ID inquiry via the network to which the MFP **10** currently belongs (the wired network or the wireless network). In this case, the device ID inquiry may include the device ID included in the NFC information. When the broadcast inquiry has been received, in the case where the device ID included in the inquiry is the device ID of the mobile device **50**, the mobile device **50** may send a response to the inquiry to the MFP **10** that is the source of the inquiry. The MFP **10** may receive the response to the inquiry via the network to which the MFP **10** currently belongs. In the case where the response is received, the determining unit **42** may determine that the current state of the MFP **10** is the state of being capable of communication with the mobile device **50** (YES in S**6** of FIG. **2**). According to the present modification, YES can be determined in S**6** of FIG. **2** in the case where the MFP **10** and the mobile device **50** are in the same subnet. In the present modification, the determination of S**6** is an example of the MFP **10** determining whether the “mobile device currently belongs to the specific network to which the communication device currently belongs”.
(210) (8) In the above embodiments, the units **40** to **46** are realized by software. However, one or more of the units **40** to **46** may be realized by hardware such as a logic circuit, etc.

Claims

1. A communication device comprising: a first wireless interface; a second wireless interface different from the first wireless interface; a processor; and a memory that stores computer-readable instructions therein, the computer-readable instructions, when executed by the processor, cause the communication device to perform: in a case where the communication device belongs to and is connected to a first wireless network formed by an access point, sending, to a terminal device via the first wireless interface, setting information of the access point, the setting information being related to SSID which is used in the terminal device for establishing, via the second wireless interface, the first wireless network including both of the communication device and the terminal device without sending a password which is used in the terminal device for establishing the first wireless network, the setting information not including the password; and after sending the setting information, executing a wireless communication of object data via the second wireless interface by using the first wireless network.
2. The communication device as in claim 1, wherein the computer-readable instructions further cause the communication device to perform: in a case where the communication device operates as a parent station, sending, to the terminal device, a wireless setting for belonging to a second wireless network formed by the communication device which operates as a parent station, the wireless setting including a password.
3. The communication device as in claim 1, wherein the computer-readable instructions further cause the communication device to perform: in a case where the communication device does not belong to the first wireless network formed by the access point, sending, to the terminal device, a wireless setting for belonging to a second wireless network formed by the communication device which operates as a parent station, the wireless setting including a password.
4. The communication device as in claim 1, wherein in a case where the terminal device currently belongs to a wireless network to which the communication device currently belongs, the setting

information and the password are not sent to the terminal device.
5. The communication device as in claim 1, wherein the communication device is any one of a printer and a scanner.
6. A communication device comprising: a first wireless interface; a second wireless interface different from the first wireless interface; a processor; and a memory that stores computer-readable instructions therein, the computer-readable instructions, when executed by the processor, cause the communication device to perform: in a case where the communication device operates as a child station and is connected to a parent station of the child station, sending, to a terminal device via the first wireless interface, setting information of the child station, the setting information being related to SSID which is used in the terminal device for establishing, via the second wireless interface, a first wireless network including both of the communication device and the terminal device without sending a password which is used in the terminal device for establishing the first wireless network, the setting information not including the password; and after sending the setting information, executing a wireless communication of object data via the second wireless interface by using the first wireless network.
7. The communication device as in claim 6, wherein the computer-readable instructions further cause the communication device to perform: in a case where the communication device operates as a parent station, sending, to the terminal device, a wireless setting for belonging to a second wireless network formed by the communication device which operates as a parent station, the wireless setting including a password.
8. The communication device as in claim 6, wherein in a case where the terminal device currently belongs to a wireless network to which the communication device currently belongs, the setting information and the password are not sent to the terminal device.
9. The communication device as in claim 6, wherein the communication device is any one of a printer and a scanner.